

SHLOK BANSAL

925-518-4010 | shlok bansal@gmail.com | [linkedin.com/in/shlok-bansal](https://www.linkedin.com/in/shlok-bansal)
Bay Area, CA | Open to relocation and travel

PROFESSIONAL SUMMARY

Biomedical Engineer with 3+ years of combined full-time and project-based medtech experience spanning implantable cardiovascular R&D, design verification, product reliability, clinical systems, and cross-functional product execution. Current R&D Engineer II at Medtronic supporting aortic-device development through fatigue, tensile, and abrasion testing, DOE/statistical analysis, fixture and test-method development, and technical evidence generation. Prior Abbott field engineering experience supporting 25+ Alinity systems adds direct customer, clinical workflow, go-live, training, and troubleshooting insight. Skilled at translating technical evidence and field feedback into clear product decisions, risk discussions, and development priorities.

CORE COMPETENCIES

- **Product Development & Cross-Functional Execution:** New product development support, stakeholder alignment, customer and field insights, clinical workflow understanding, technical presentations, training, go-live support, schedule and priority coordination
- **Verification, Testing & Reliability:** Fatigue, tensile, abrasion, burst and benchtop testing; test-method, fixture and setup development; equipment qualification/IQ; anomaly investigation; root-cause and CAPA support
- **Data Analysis & Decision Support:** DOE, P-diagrams, ANOVA, Minitab, MATLAB, trend analysis, failure-behavior analysis, evidence synthesis, structured data and imaging review
- **Documentation, Quality & Tools:** SOPs, test summaries, verification documentation, DHF support, IEC 60601, ISO 5840, FDA/ISO-regulated environments; SolidWorks, Python, Arduino, Excel, PowerPoint, Word

PROFESSIONAL EXPERIENCE

R&D Engineer II | Medtronic CardioVascular Surgery | Santa Rosa, CA

October 2025 - Present

- Support development and verification testing for implantable aortic-device components through fatigue, tensile, abrasion, and other benchtop evaluations.
- Develop and refine test methods, fixtures, boundary conditions, instrumentation, and measurement approaches using DOE and P-diagram-based problem structuring to improve repeatability, test fidelity, and confidence in performance evaluation.
- Execute installation qualification protocols on Bose ElectroForce fatigue systems and document setup verification, equipment readiness, and regulated-lab test capability.
- Analyze soft-goods and long-duration test data using Minitab and ANOVA; investigate anomalies, wear patterns, environmental drift, and potential failure mechanisms to guide engineering next steps.
- Standardize suture and textile tensile/abrasion workflows, coordinate sterilization and material-readiness activities with external vendors, and align instrumentation and procedures for consistent reporting and faster design comparisons.
- Build structured analytics workflows across test data, imaging, documents, and video, and translate findings into concise reviews supporting R&D, Quality, laboratory, product-risk, and program-priority decisions.
- Coordinate concurrent test readiness, schedules, execution priorities, and next-step actions across engineering and laboratory teams.

Field Service Engineer | Abbott Core Diagnostics | SF Bay Area / Region West

September 2023 - October 2024

- Deployed, qualified, validated, and supported 25+ Alinity diagnostic systems across hospital and reference laboratories through instrument checks, field verification, QC workflows, software updates, and system-readiness activities.
- Led go-live readiness, product demonstrations, customer training, and workflow support while partnering directly with clinical users, lab leadership, Quality, technical support, and field leadership.
- Diagnosed hardware, software, workflow, and environmental failures; supported root-cause and CAPA-related actions that contributed to a 15% improvement in instrument uptime and reduced repeat escalations.
- Translated recurring field issues and customer feedback into escalation summaries, troubleshooting guidance, service bulletins, and improved documentation to strengthen knowledge transfer and onboarding.
- Managed competing installation and service priorities across a regional account base, coordinating schedules, issue resolution, and stakeholder communication in fast-paced clinical environments.

ADDITIONAL MEDTECH EXPERIENCE

EEG Stroke Detection Senior Capstone | Medtronic Neurovascular | *Irvine, CA* September 2022 - June 2023

- Developed and validated a MATLAB-based EEG stroke-classification model using DAR, pdBSI, and PSD features, reducing false positives by 40% through SOBI and ICA artifact filtering.
- Designed and tested experimental protocols to evaluate system functionality and supported cross-functional optimization of an early-stage prototype for preclinical evaluation.
- Drafted Class II verification protocols and aligned algorithm outputs with early-stage system-integration and diagnostic-design needs for a portable neurodiagnostic concept.
- Presented findings to R&D and clinical stakeholders, translating signal-processing performance into verification recommendations and future system-development implications.

R&D Intern | Foldax | *Irvine, CA* October 2022 - February 2023

- Executed ISO 5840-compliant fatigue, burst, and tensile testing on polymeric heart valves to support device-performance evaluation and design iteration.
- Redesigned SolidWorks fixtures and refined benchtop setups, improving test repeatability by 15% and increasing confidence in mechanical data.
- Conducted feasibility testing on materials and mechanical concepts, documented design iterations and protocols, and partnered with vendors on material sourcing and compatibility evaluation.
- Analyzed test results with R&D and Quality teams to inform valve-design changes, material decisions, and next-phase development activities.

System Integration Intern | Medtronic Neuromodulation | *Minneapolis, MN* June 2022 - August 2022

- Designed and prototyped Arduino-based electromechanical test systems using sensors, 3D-printed components, and precision motors to support recharge validation for implantable neurostimulators.
- Developed automation workflows and experimental setups that improved product-testing efficiency by 20% while supporting mechanical and electromechanical system integration.
- Authored SOPs and contributed to IEC 60601-aligned test protocols and verification documentation included in the design history file.
- Identified and documented hardware, sensing, fixture, and system-integration issues; collaborated on corrective actions and verified implemented changes to improve repeatability and compliance.

LEADERSHIP EXPERIENCE

Vice President, External | Biomedical Engineering Society at UCI | *Irvine, CA* September 2022 - June 2023

- Managed relationships with industry representatives, alumni, engineering organizations, and BMES chapters to expand cross-campus and industry engagement.
- Planned national virtual events for 60+ attendees and coordinated engineering outreach for high-school students in underserved communities.

PROFESSIONAL DEVELOPMENT

- Medtronic EVAR Academy 101-202: aortic anatomy, endovascular procedures, device concepts, and the clinical landscape.

EDUCATION

Bachelor of Science in Biomedical Engineering | University of California, Irvine | *Irvine, CA* June 2023